Statistical learning project

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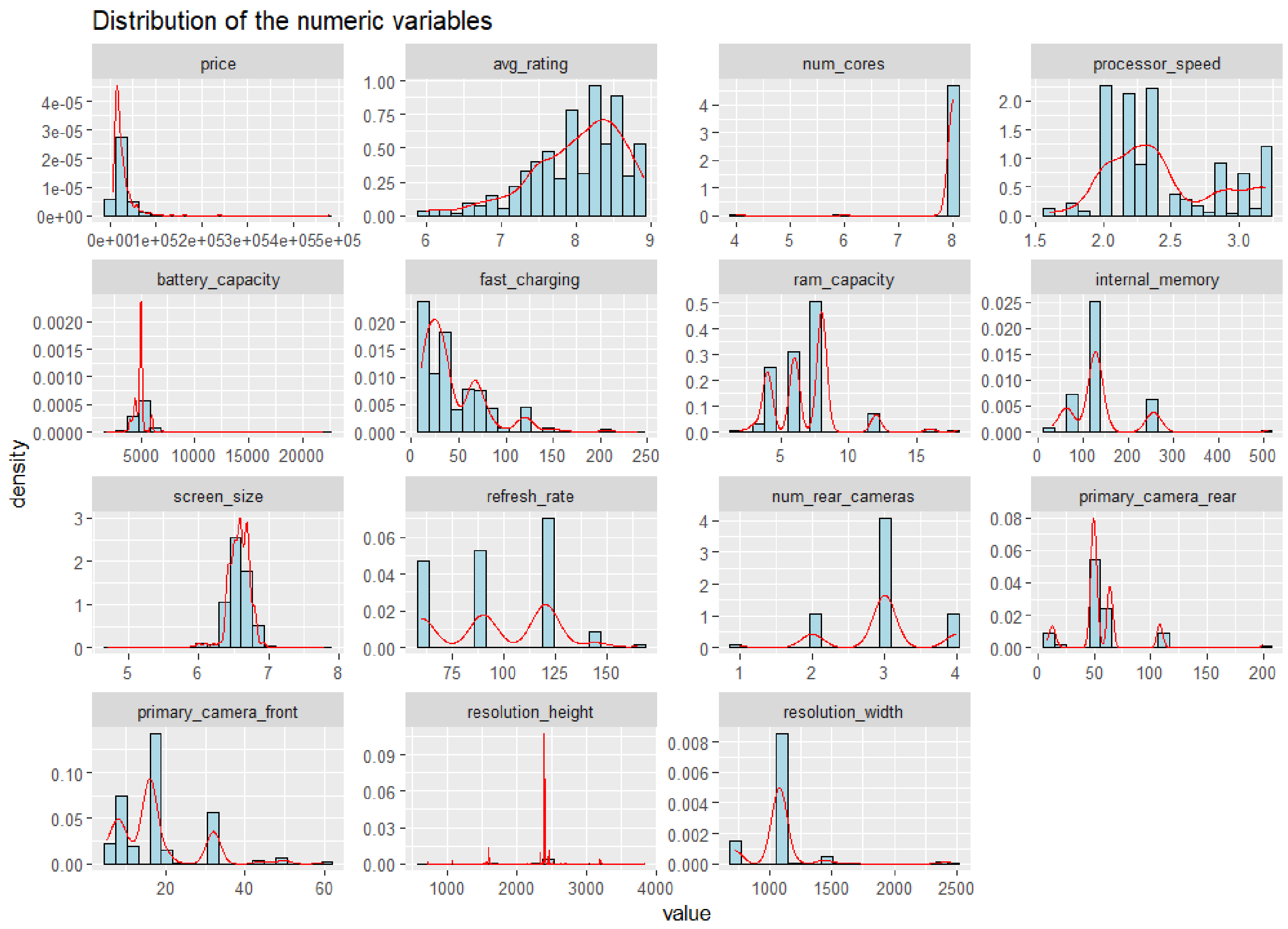
Unsupervised learning

Smartphones Dataset

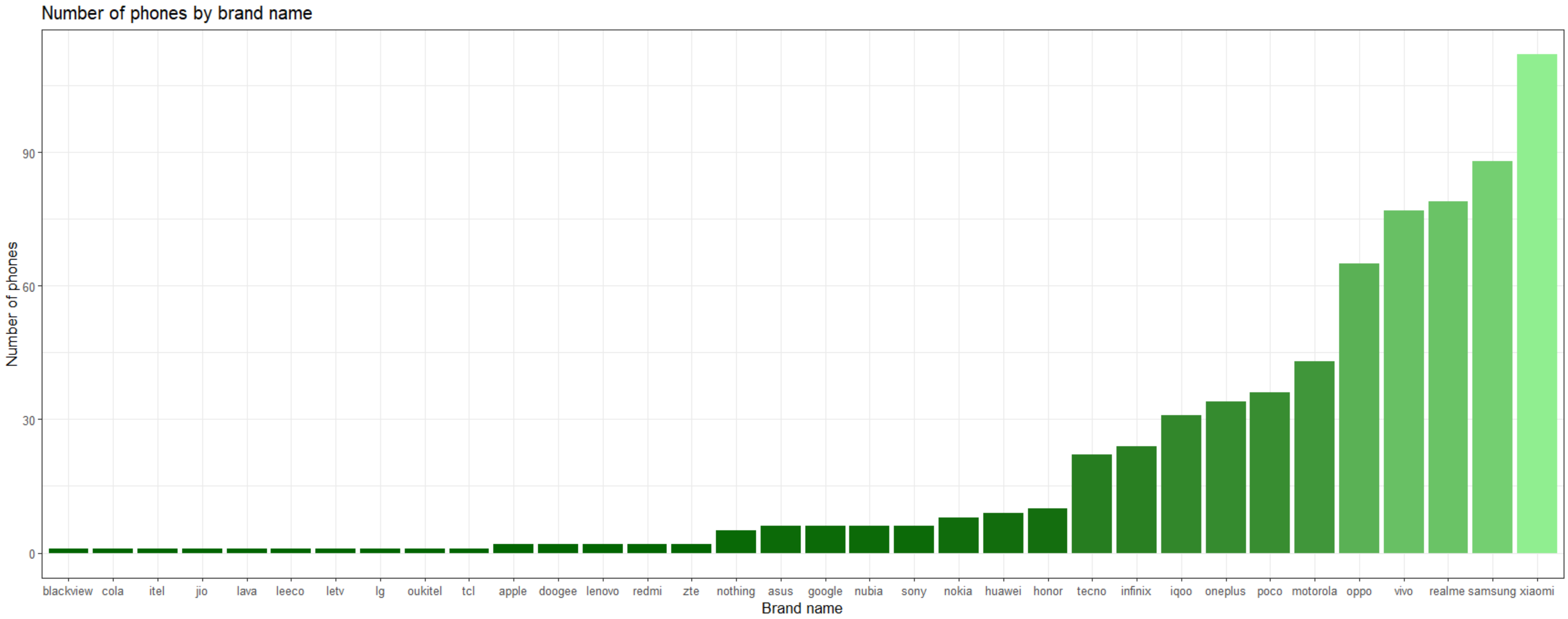
For this part of the project, a dataset about smartphones was choosen which contains 980 observations of 22 variables. The dataset included different smartphones features such as:

- *Battery capacity*
- *Ram capacity*
- *Screen size*
- *Processor speed*
- *Price and so on*

Exploratory data analysis (EDA)

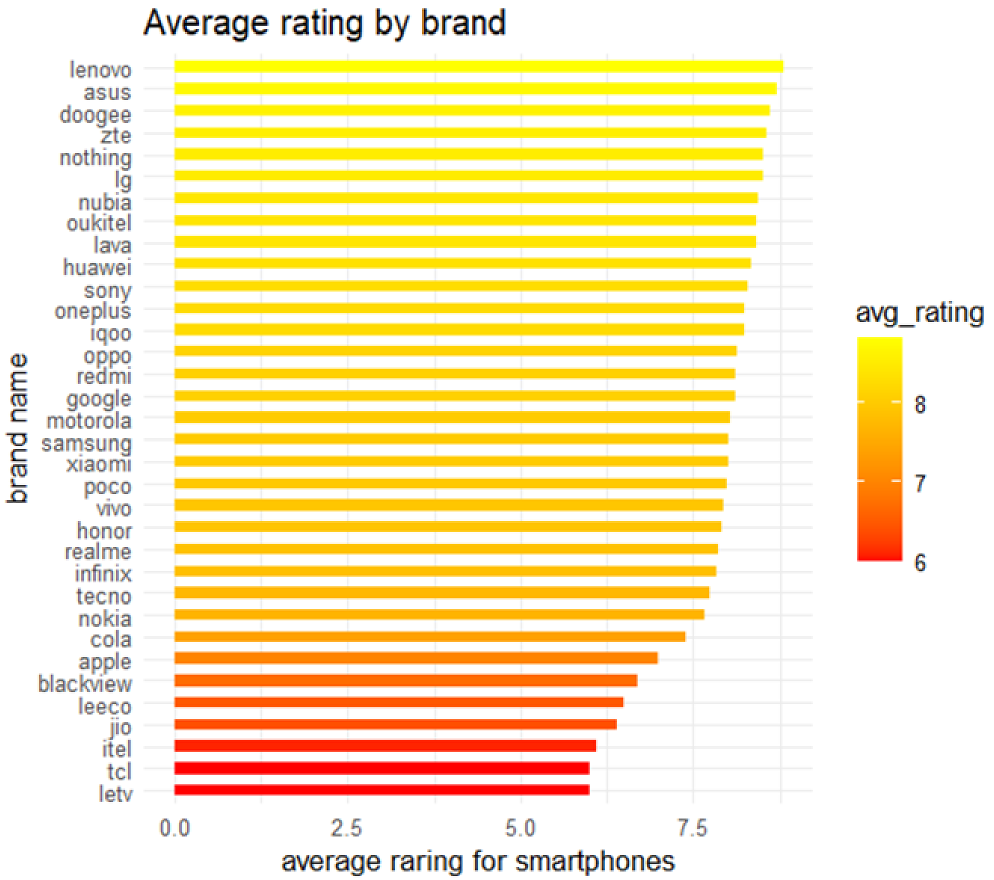


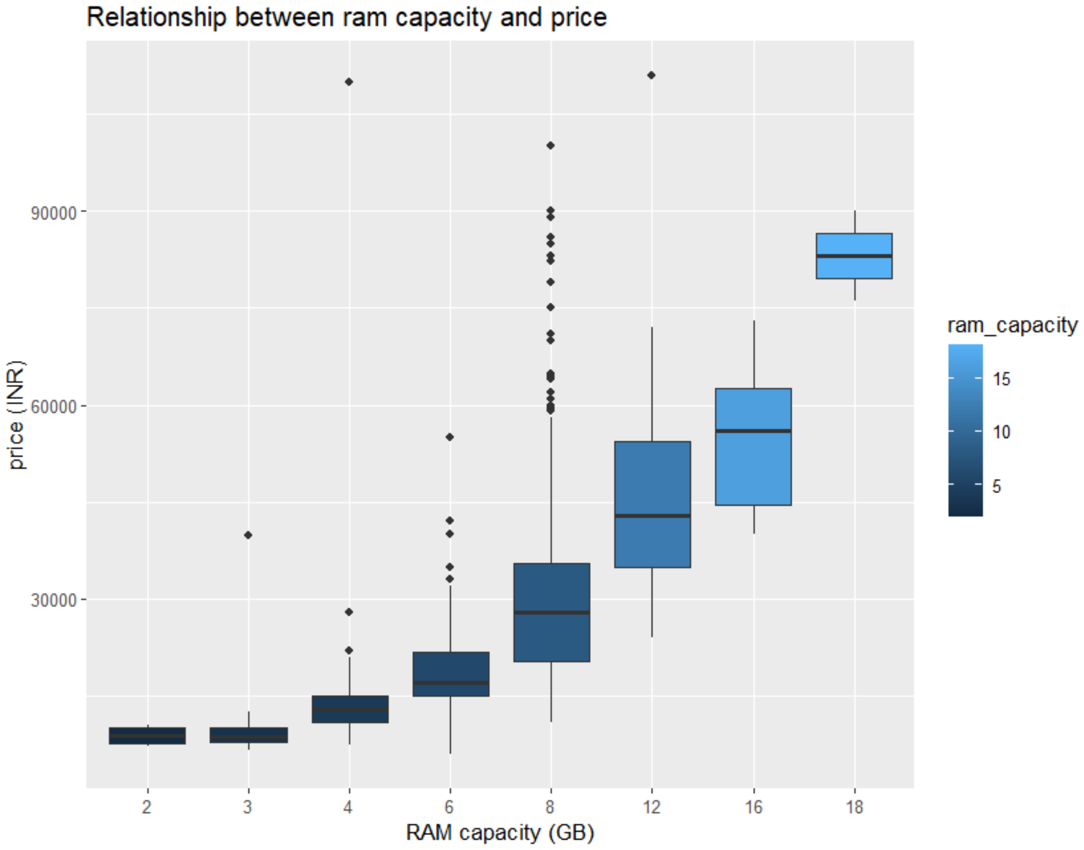
Exploratory data analysis (EDA)



Xiaomi, Samsung and Realme are the top three brands in terms of the number of smartphones counted in the dataset that belongs to them. The other brands have a significantly smaller representation in the datset.

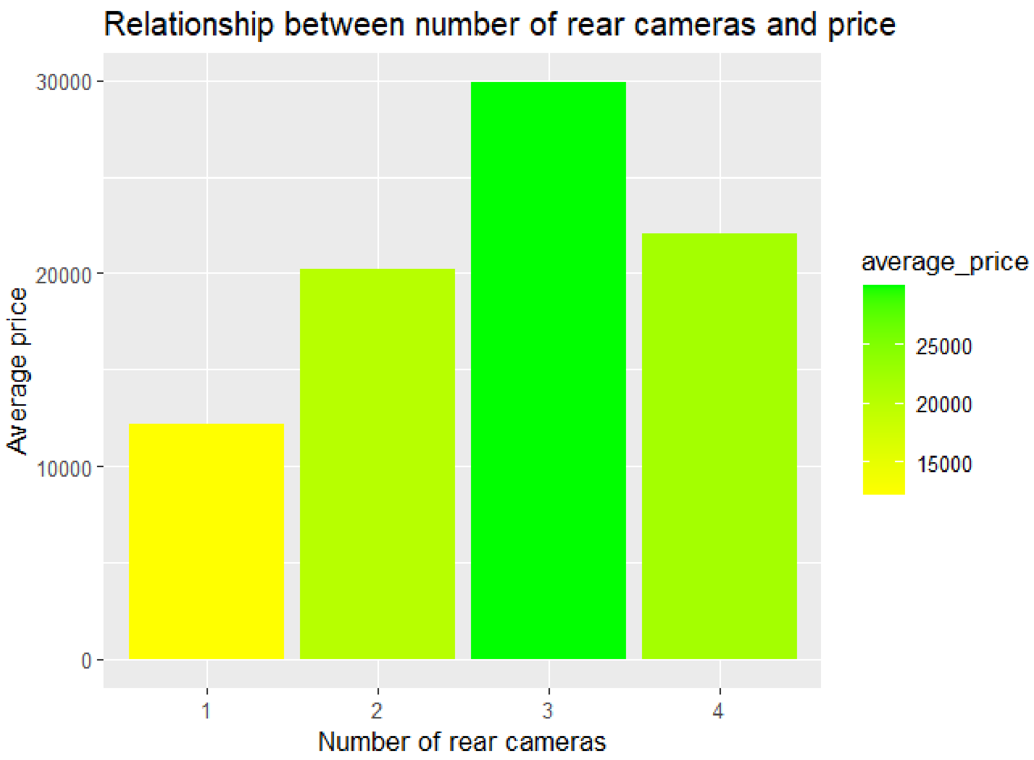
Exploratory data analysis (EDA)

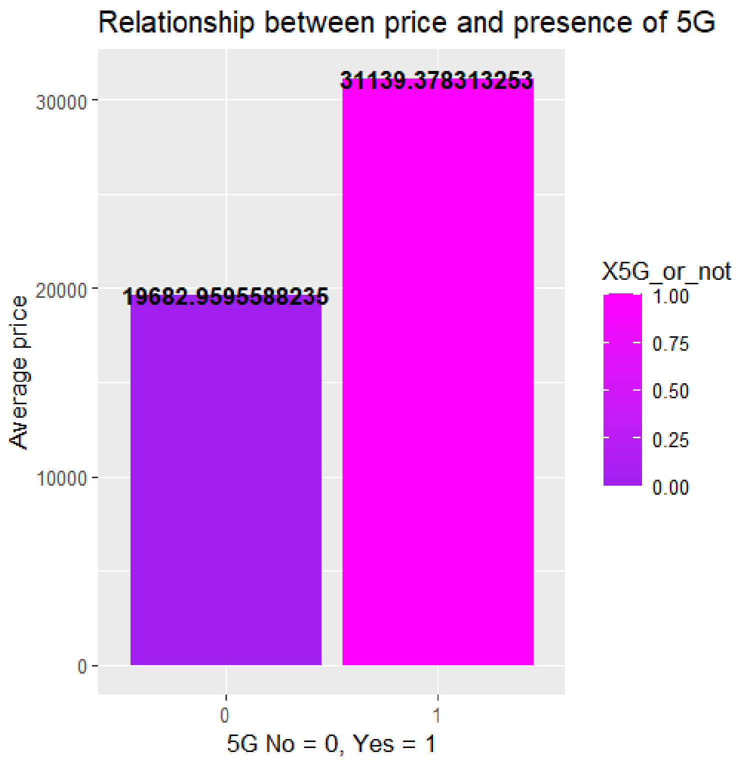




- *This bar plot highlights how consumer satisfaction varies by brand and highlights the most popular smartphone's brand based on the average rating.*
- *The RAM capacity has a positive correlation with the prices of the smartphones, in fact, as the RAM capacity increases, also the price of the smartphones increases*

Exploratory data analysis (EDA)

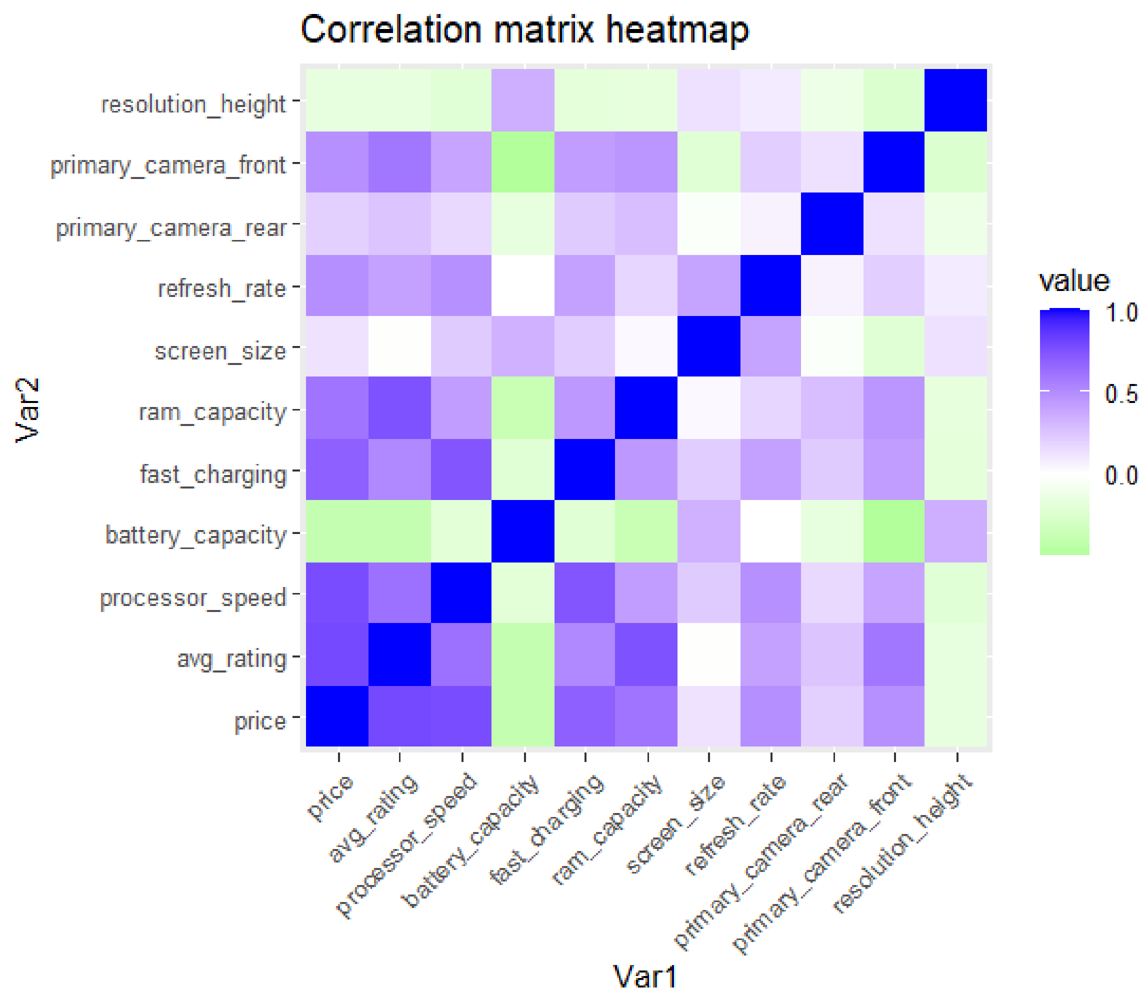




- *Smartphones with 3 rear cameras show the highest average price, suggesting that this is a feature that higly influenze the price of the devices. However, this relationship is not perfectly linear since smartphones with 4 rear cameras have lower average price than the ones with only 3 cameras.*

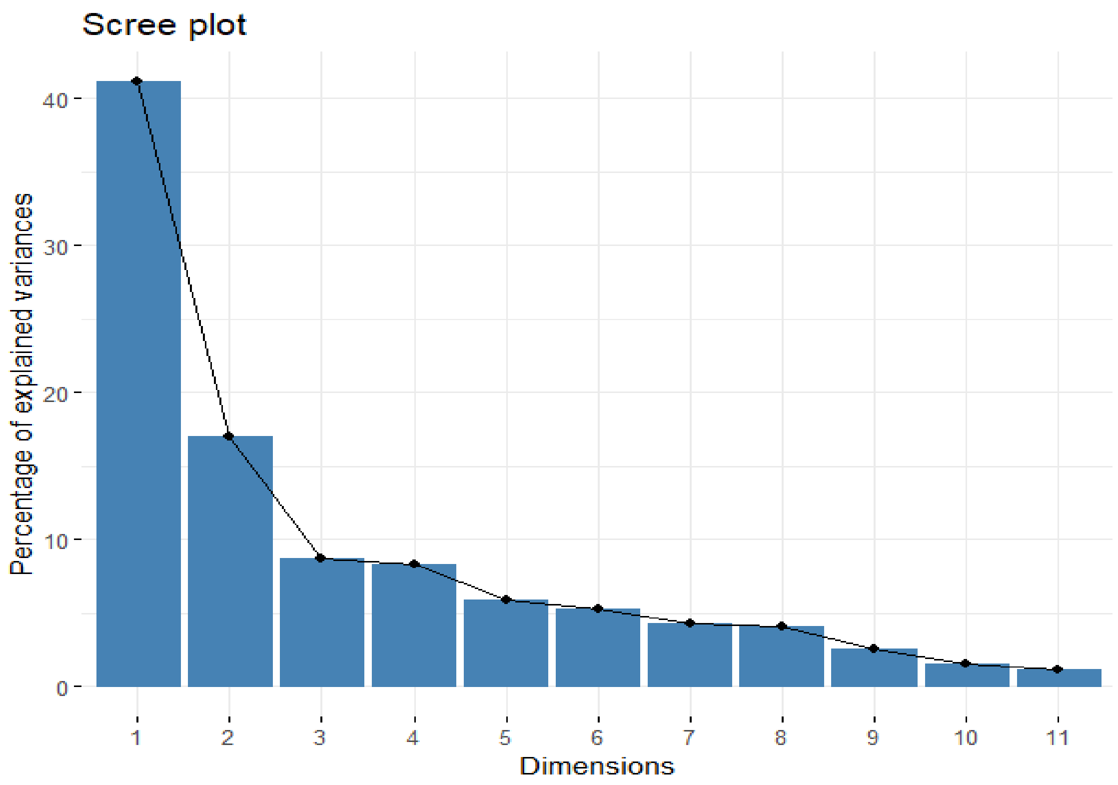
- *This suggests that the 5G is tipically present in more premium smartphone models since the average price increases if there is the 5G implemented on the smartphone.*

Exploratory data analysis (EDA)



The heatmap of the correlation matrix shows the strength of the relationships between the numerical features of the dataset. Strongest positive correlations with price: RAM capacity, processor speed, and screen size. Overall, most of the variables show moderate correlation with each other. This evaluation is important for identifying key variables for modelling and feature selection.

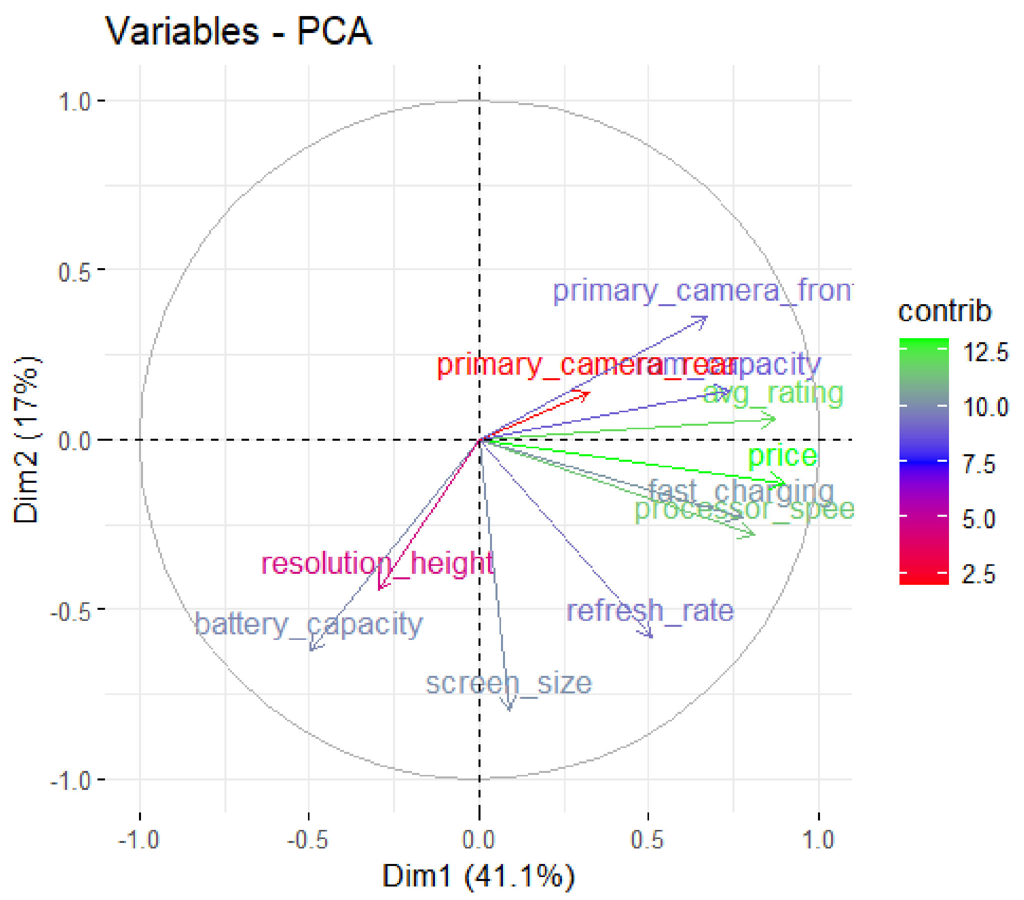
Principal component analysis (PCA)



- *Below there are the results of the principal componet analysis (PCA). The scree plot on the left is used to represent the portion of the variance explained by each principal component resulting from the PCA analysis.*

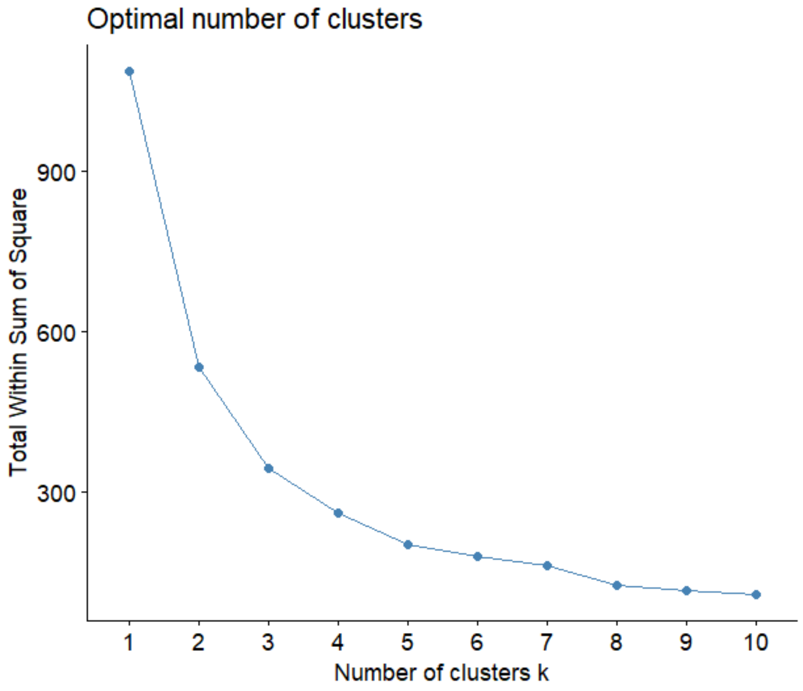
```
> summary(pca_results)
Importance of components:
                          PC1    PC2     PC3    PC4     PC5     PC6     PC7     PC8     PC9    PC10    PC11
Standard deviation     2.1268 1.3695 0.97776 0.9549 0.80530 0.76177 0.68742 0.67299 0.52681 0.41908 0.35486
Proportion of Variance 0.4112 0.1705 0.08691 0.0829 0.05896 0.05275 0.04296 0.04117 0.02523 0.01597 0.01145
Cumulative Proportion  0.4112 0.5817 0.66862 0.7515 0.81047 0.86322 0.90618 0.94736 0.97259 0.98855 1.00000
```

Variable contribution plot

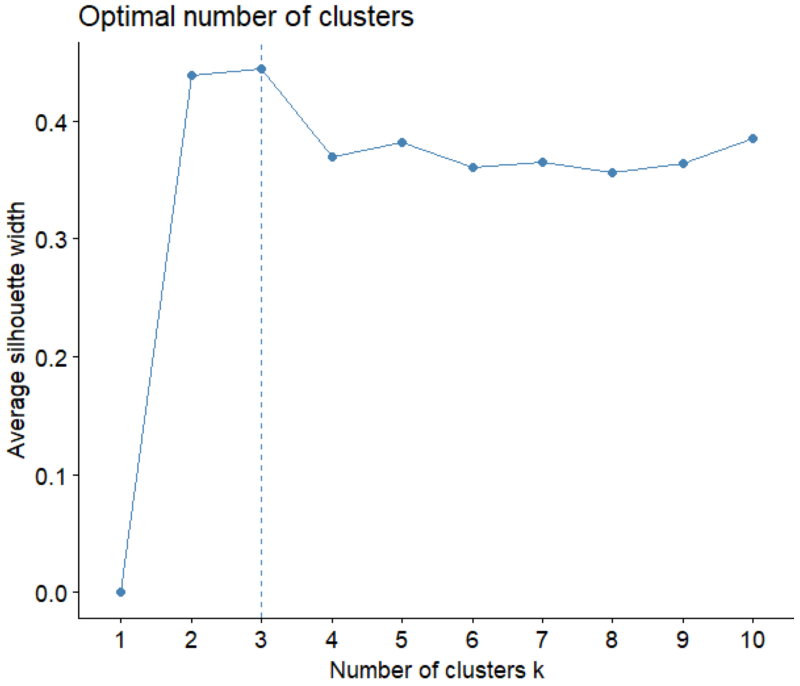


Clustering

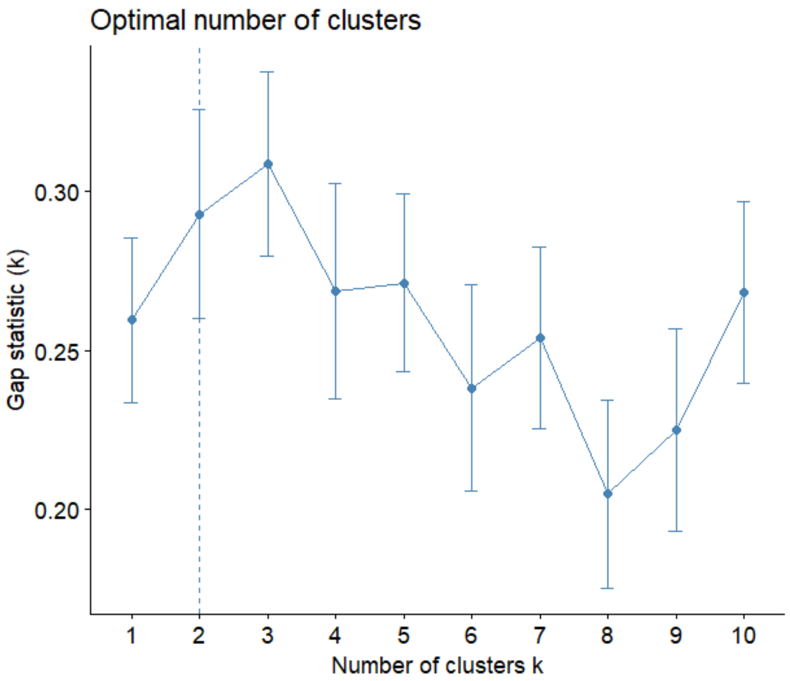
WSS (Within sum of squares) method or "elbow method"



Silhouette method:



Gap method

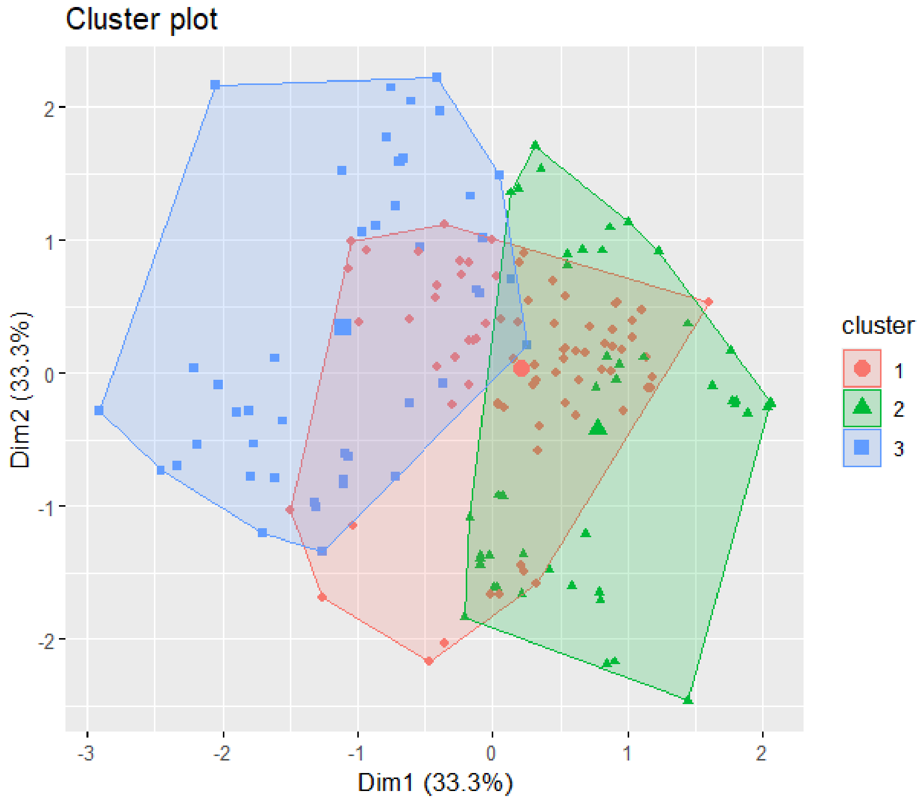


From this analysis, the optimal number of clusters to retain should be between k = 2 and k = 3

K - means

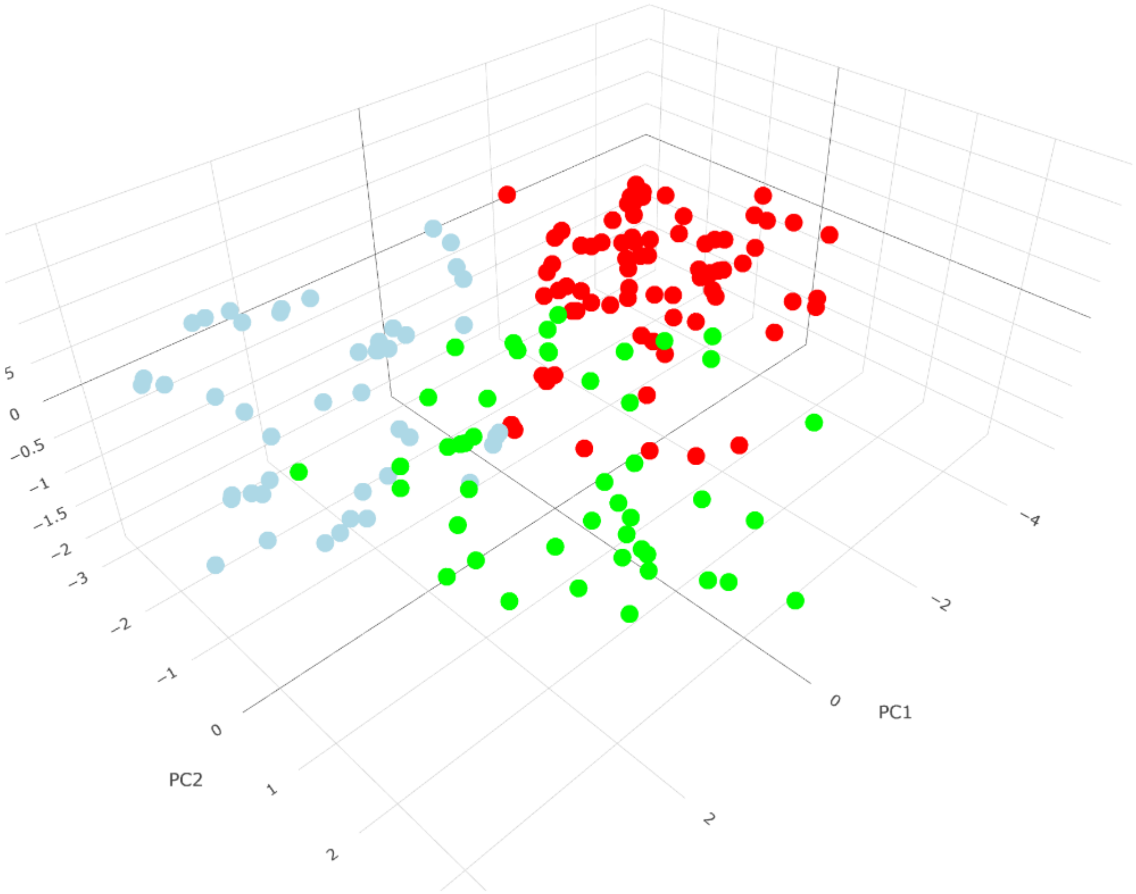
Classification results with k = 2

Classification results with k = 3



The K-means algorithm was applied with k = 2 and k = 3 to explore natural groupings. While k = 2 offers a clear separation, k = 3 provides more detailed segmentation, although with some overlap.

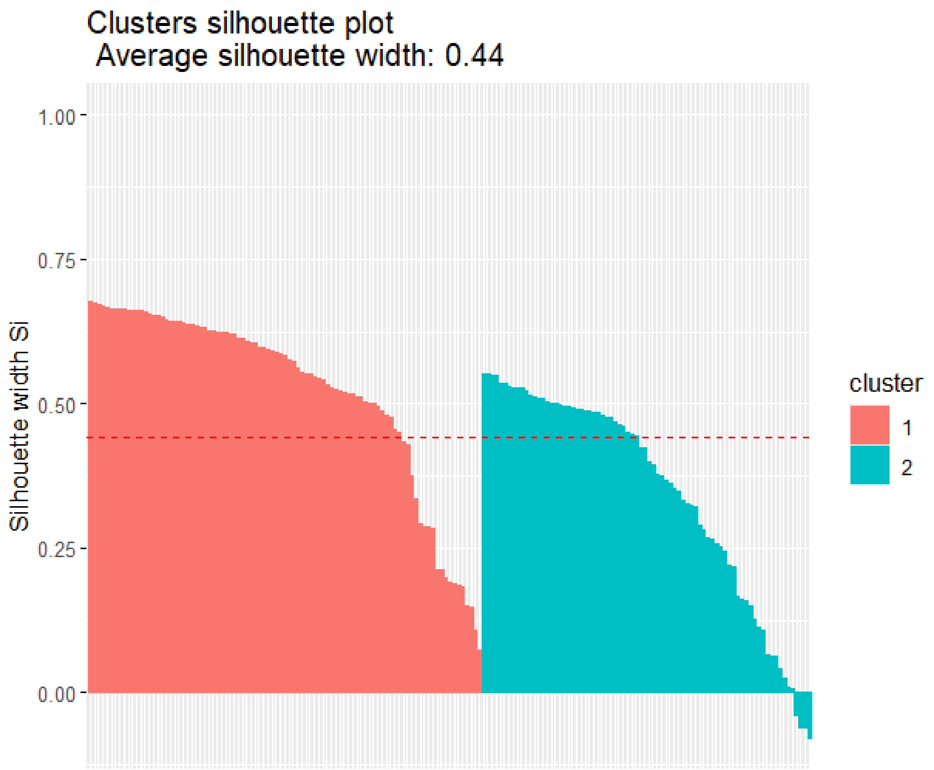
3D visualization of K – means with k = 3



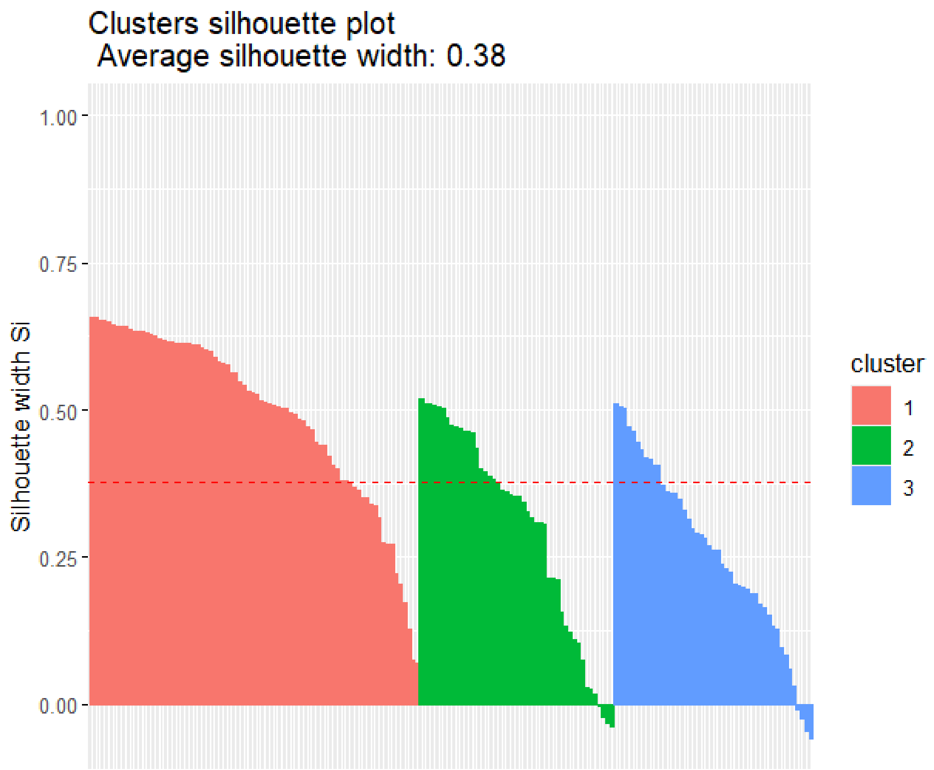
- *Here's a clearer visualization of the clusters separation for the case where k = 3 in a 3D form.*

K – means silhouette score

K = 2

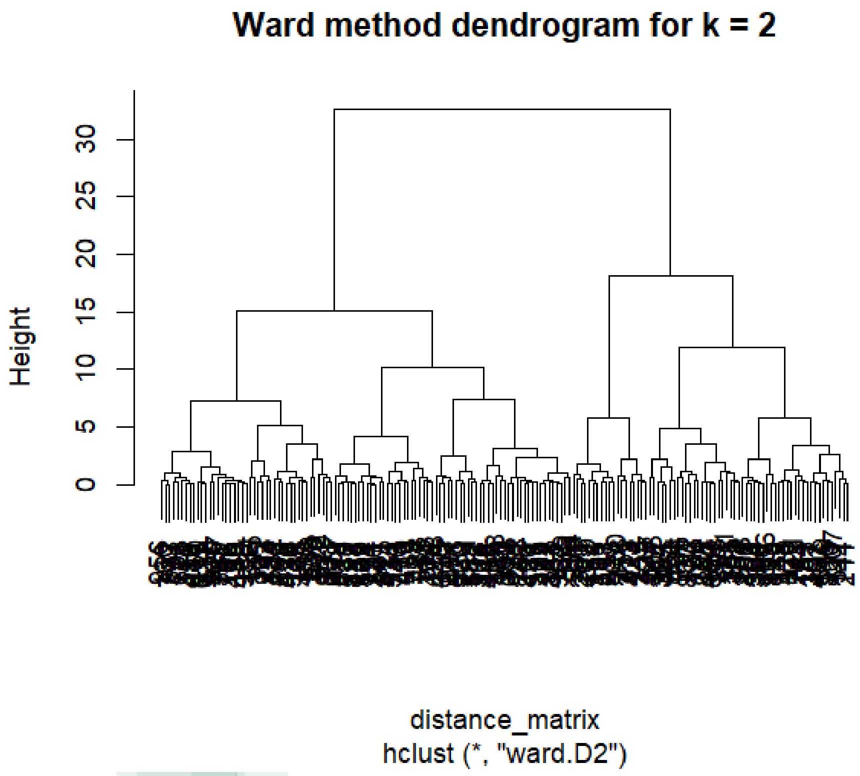


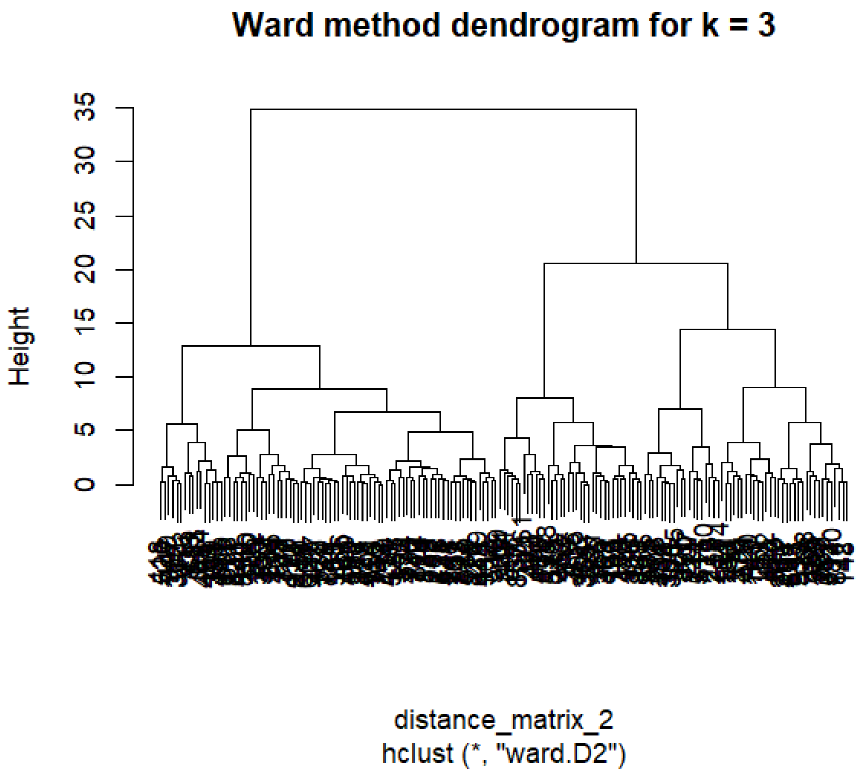
K = 3



We can see that for k = 2 we have a higher silhouette score, so the optimal number of clusters to be considered should be k = 2.

Hierarchical clustering





Results

- For PCA, the optimal numer of principal components, which explain most of the vairance, is two
- The optimal number of clusters is two
- K – means with k = 2 is the most effective clustering method for this dataset

Supervised learning

Breast cancer Dataset

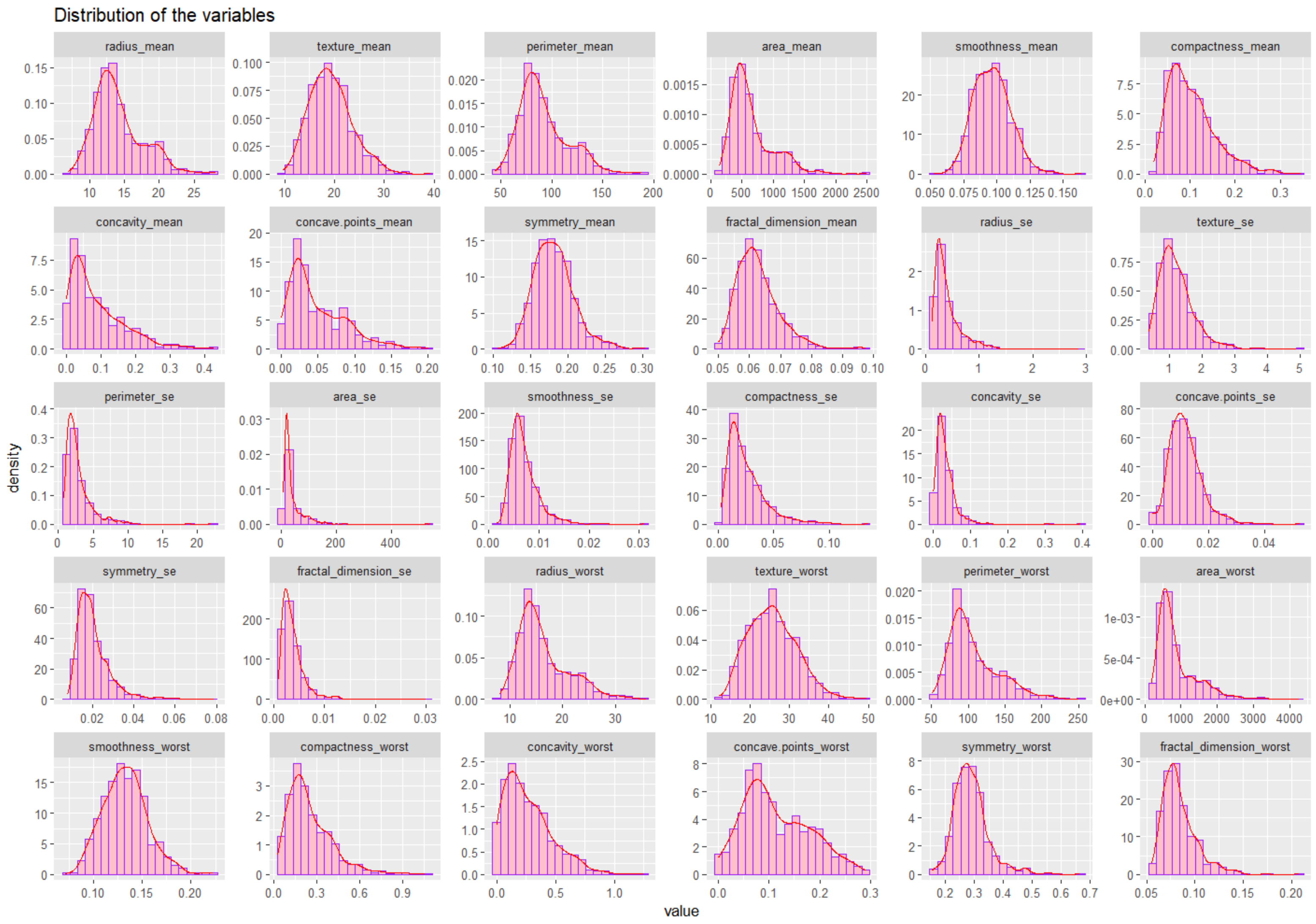
For this part of the project, a breast cancer dataset was chosen. This dataset contains ***569 observations*** *and* ***31 variables*** *about various biological characteristics of cell nuclei derived from breast cancer mass images. Among them we have:*

- ***Diagnosis:*** *it's the target variable and it can be equal to M (malignant) and B (benign)*

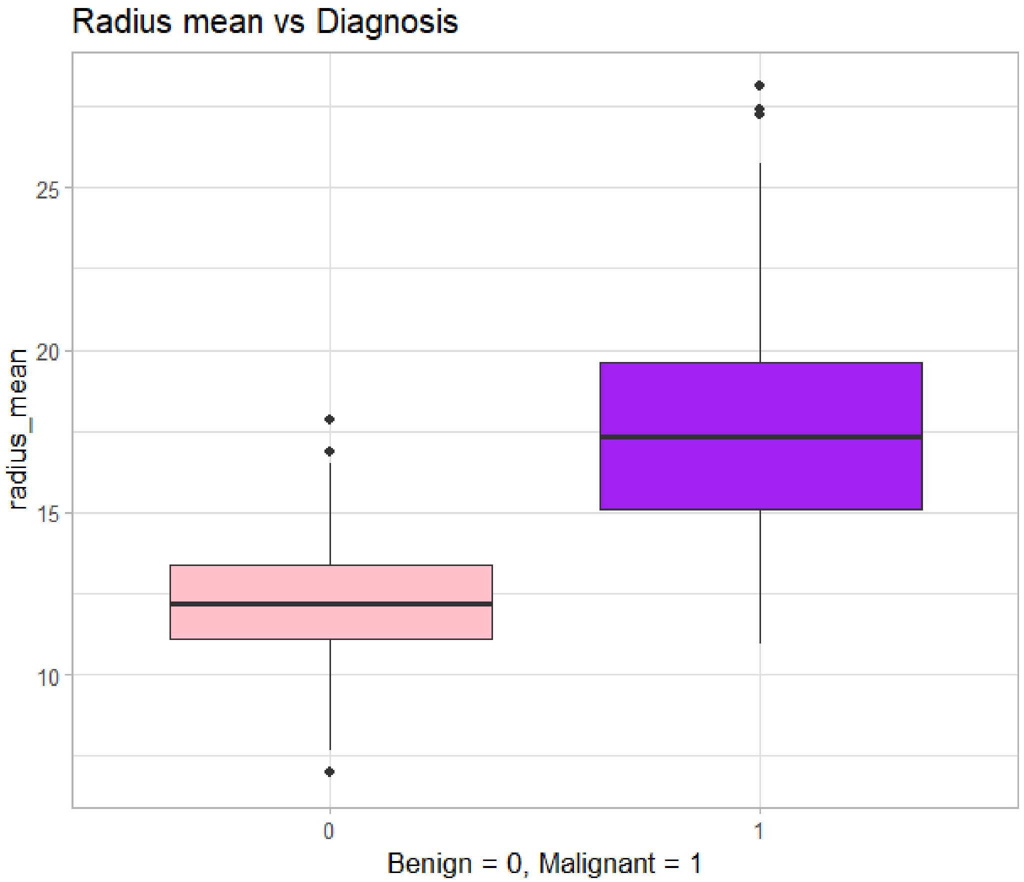
The remaining 30 numerical features are grouped in tree sets:

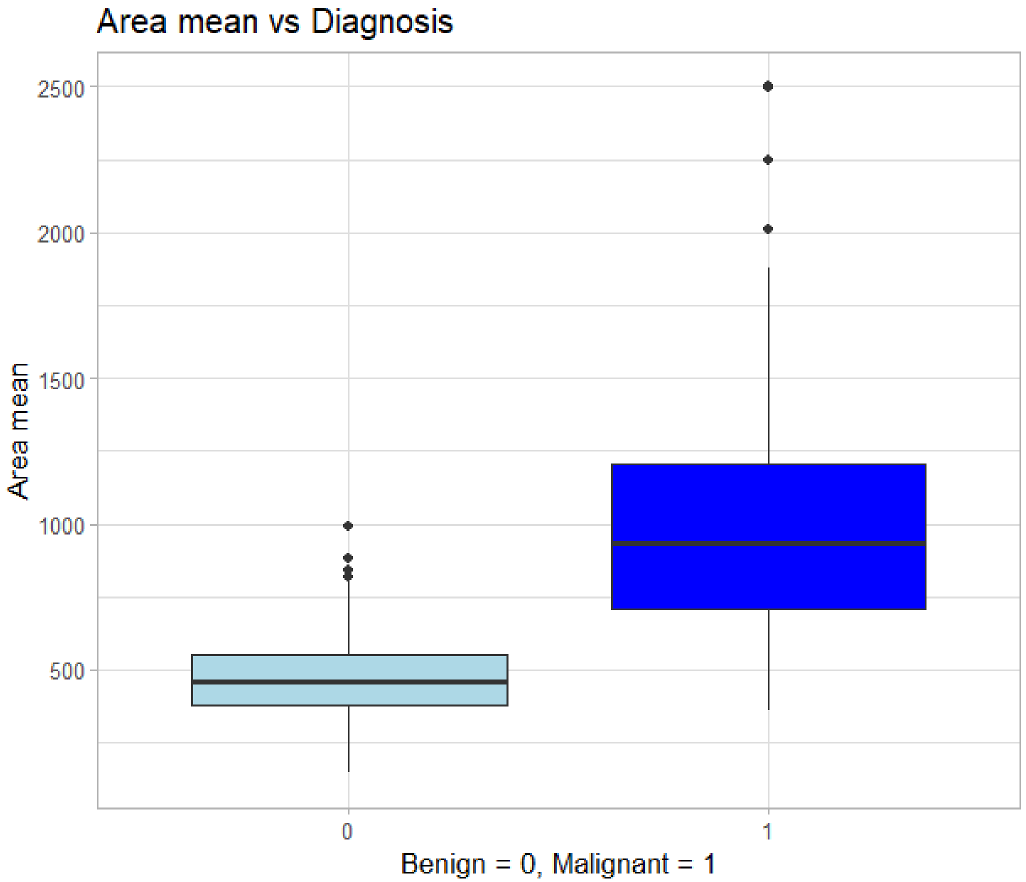
- ***Mean values:*** *like texture_mean, perimeter_mean, area_mean and radius_mean.*
- ***Standard error values:*** *like smoothness_se, compactness_se and symmetry_se.*
- *Worst case values: like radius_worst, concavity_worst and fractal_dimension_worst.*

Exploratory data analysis (EDA)



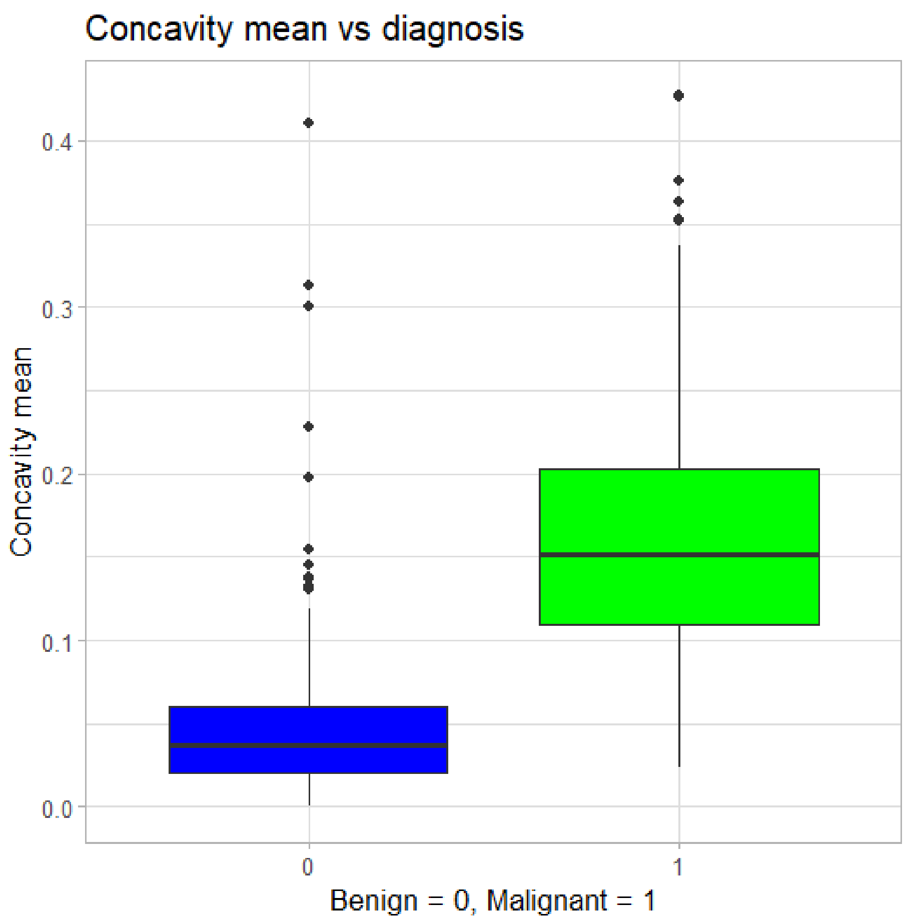
Exploratory data analysis (EDA)

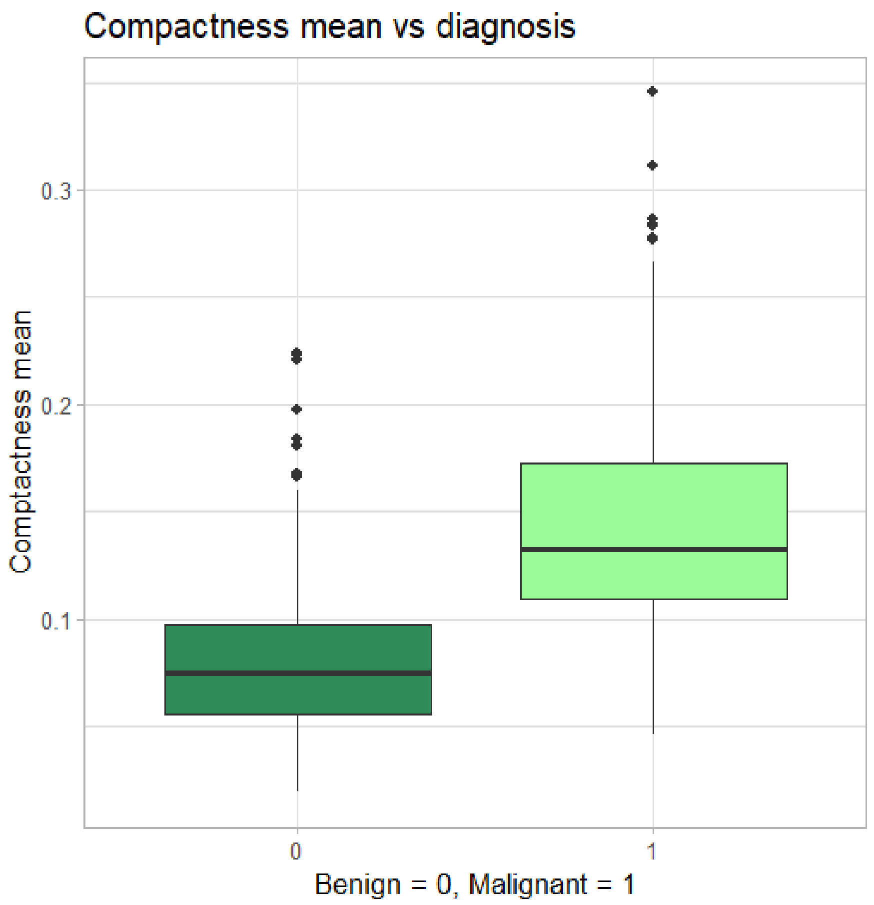




- *Higher values of radius mean are strongly associated with the diagnosis of a malignant tumour*
- *Malignant tumours tend to occupy a larger area , making area mean a key diagnostic feature*

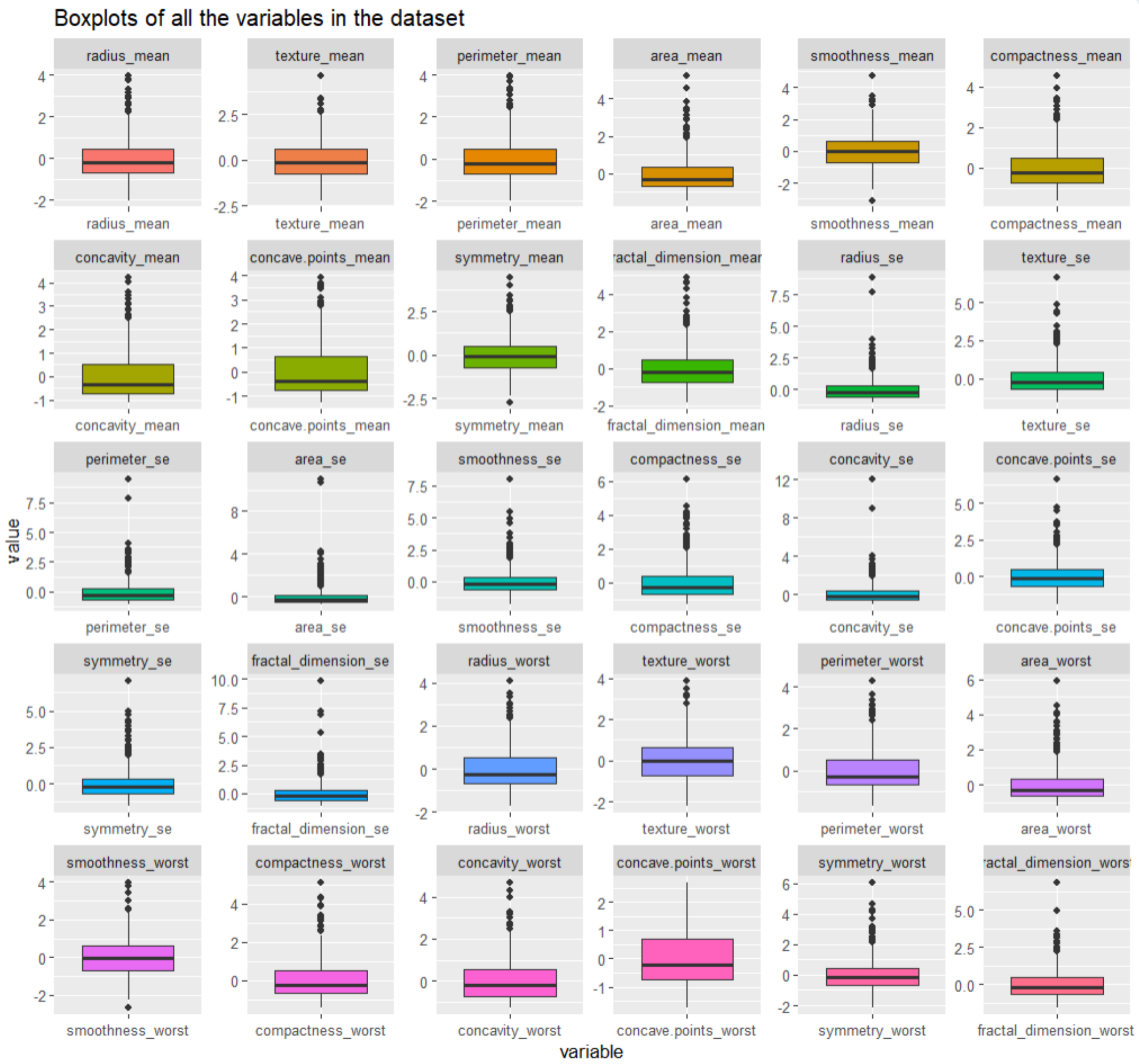
Exploratory data analysis (EDA)



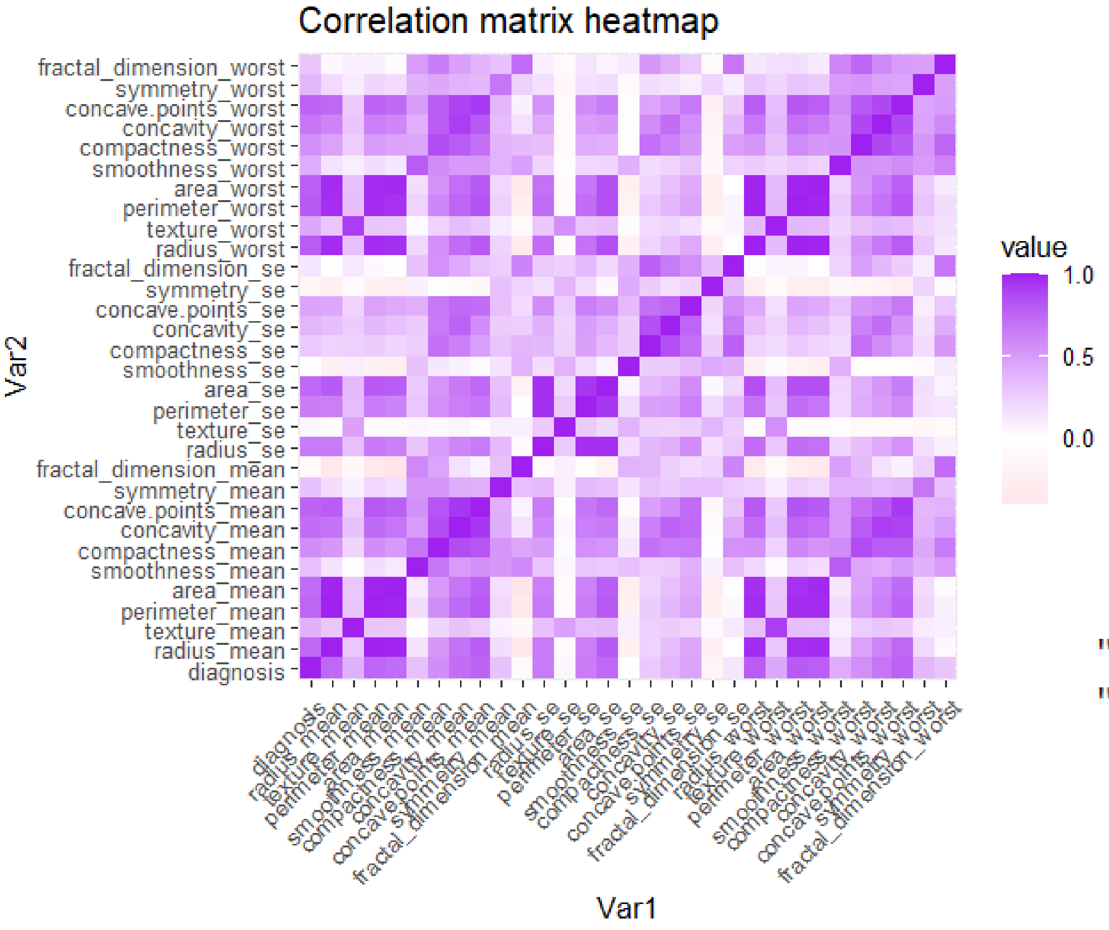


- *Increased concavity mean indicates a more aggressive and irregular growth, which are two factors that influence the most the malignant diagnosis in the patients.*
- *Malignant tumours show higher values of compactness, indicating a denser and more irregular growth patterns.*

Exploratory data analysis (EDA)



Exploratory data analysis (EDA)

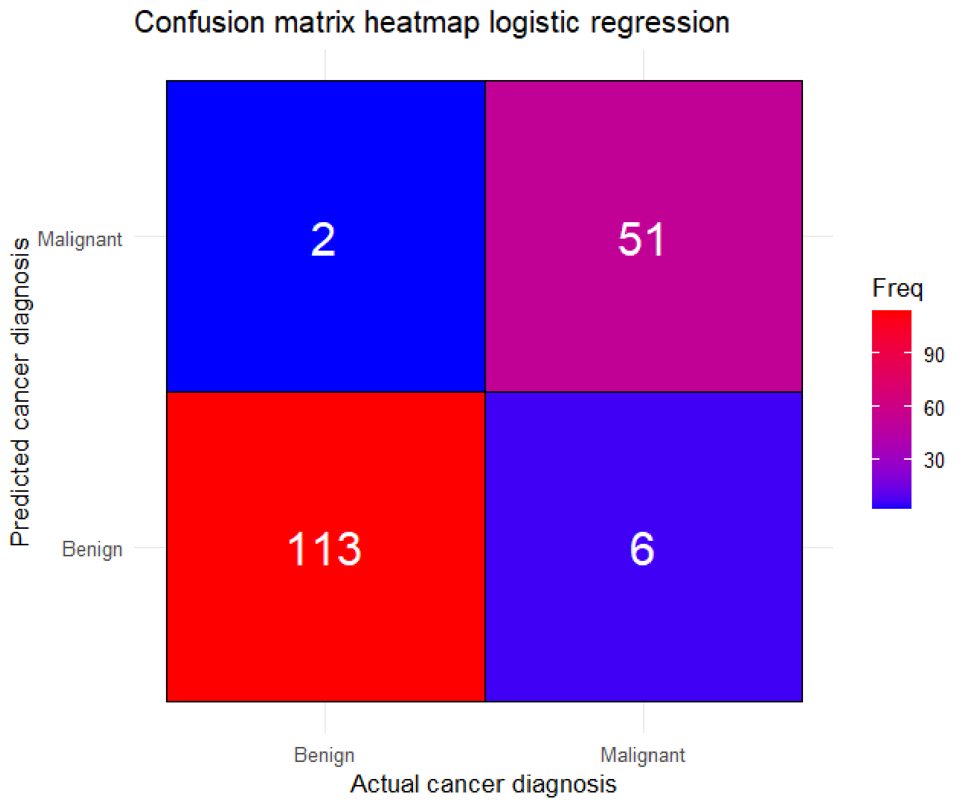


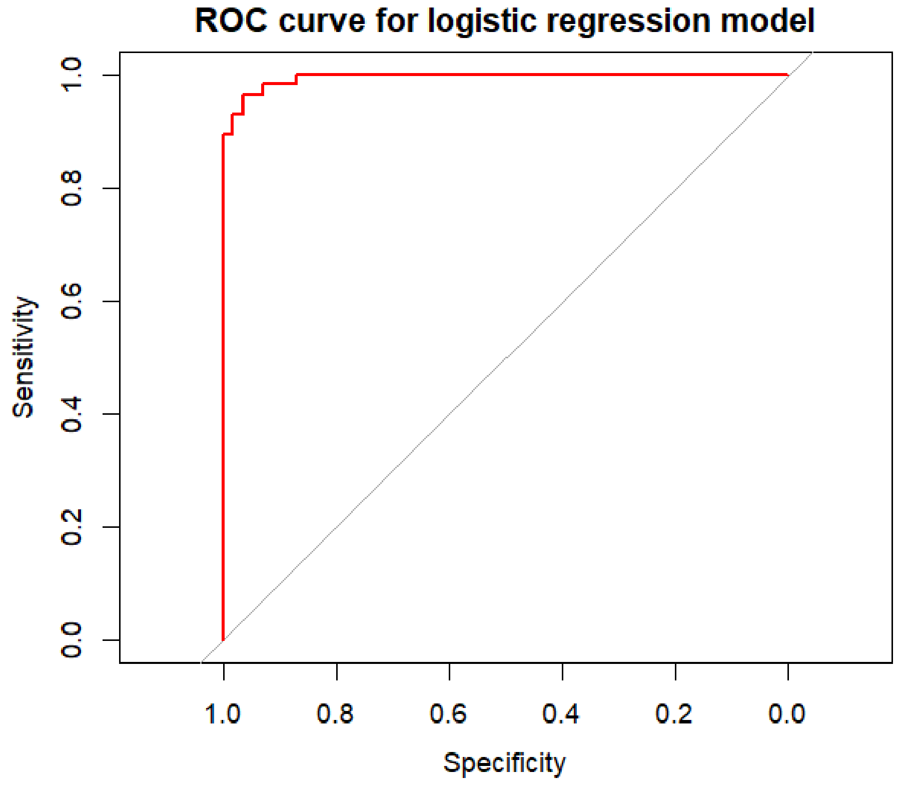
The heatmap reveals strong correlations between several features. The ones represented below are the highest correlated variables which were removed from the dataset to avoid redundancy and potential bias in the learning algorithms used throught the analysis.

```
"concavity_mean"      "concave.points_mean" "perimeter_worst"
"perimeter_mean"      "area_se"             "radius_mean"

"radius_worst"        "area_worst"
"perimeter_se"        "texture_mean"
```

Logistic regression

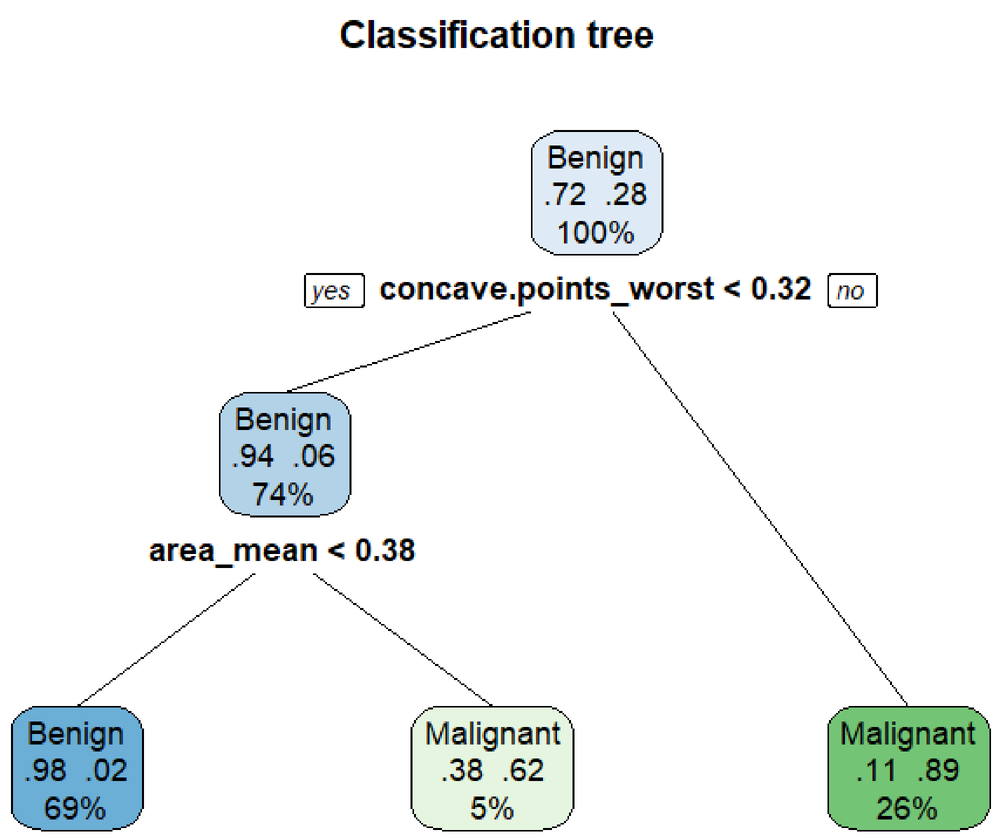




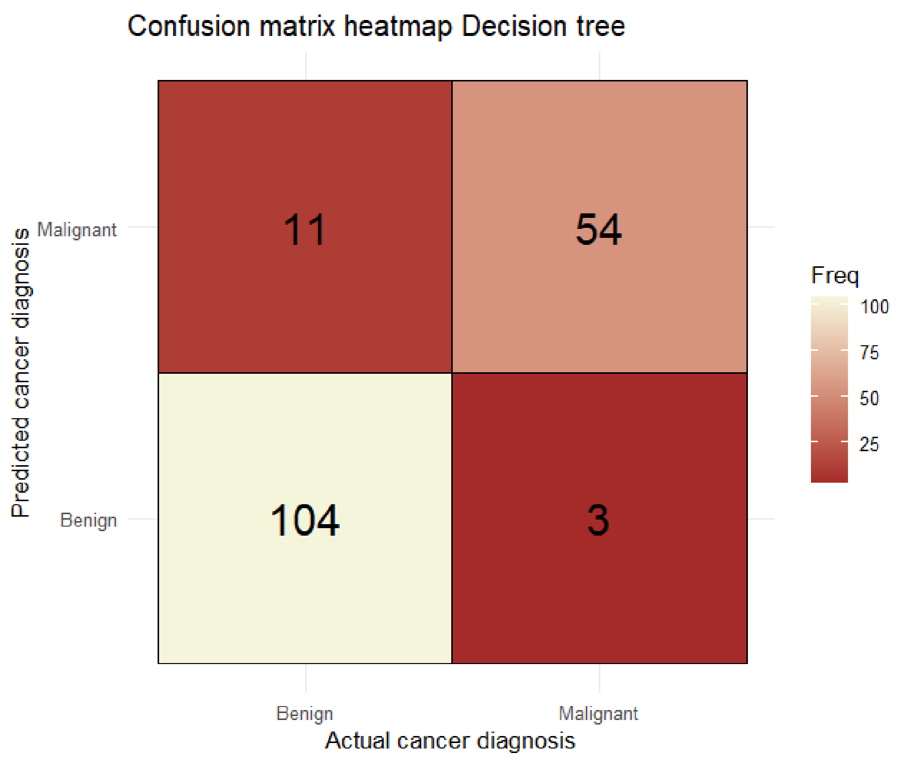
"Logistic regression accuracy: 0.9535"

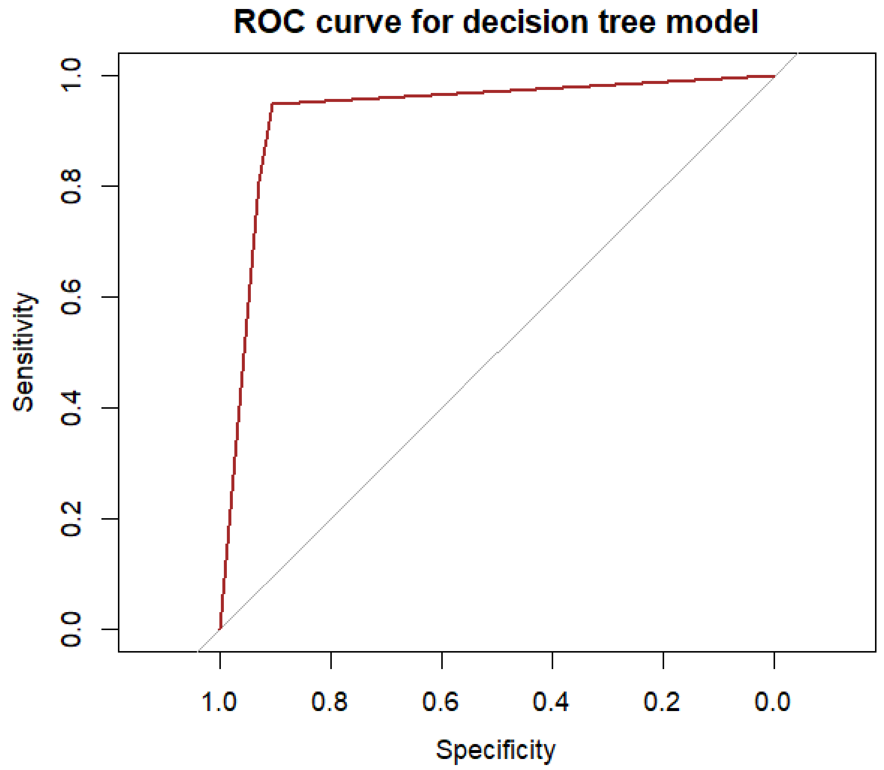
Area under the curve: 0.9947

Decision tree



Decision tree

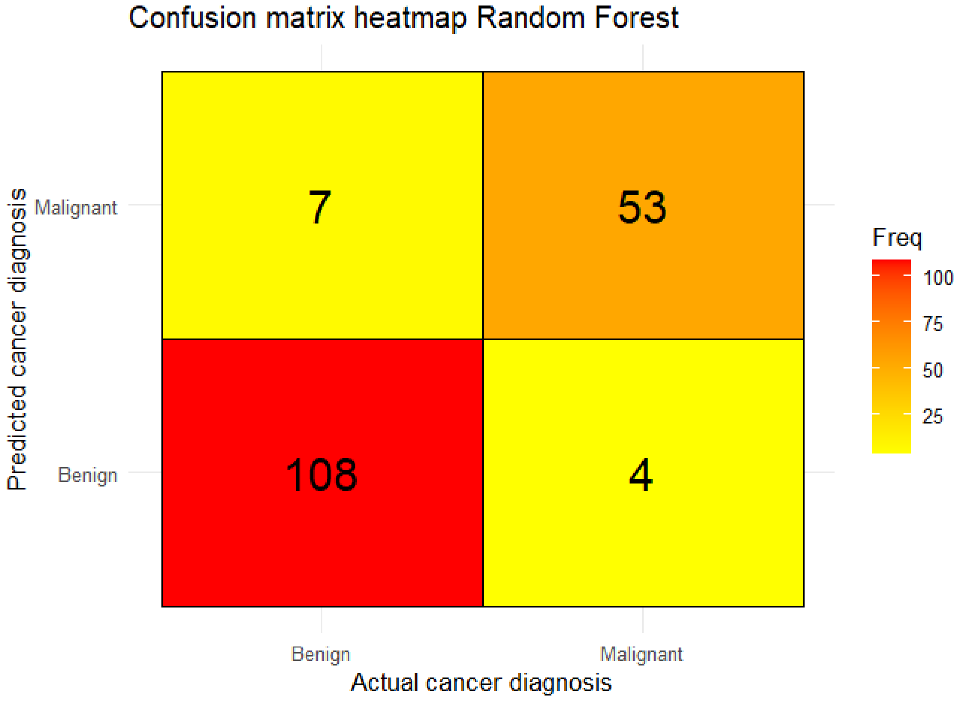


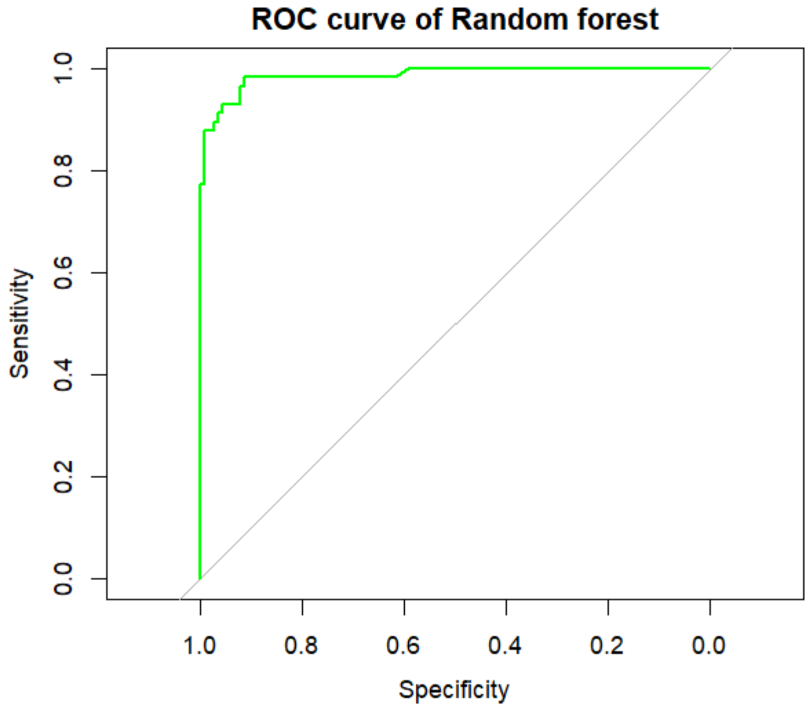


"Decision tree accuracy: 0.9186"

Area under the curve: 0.9315

Random Forest



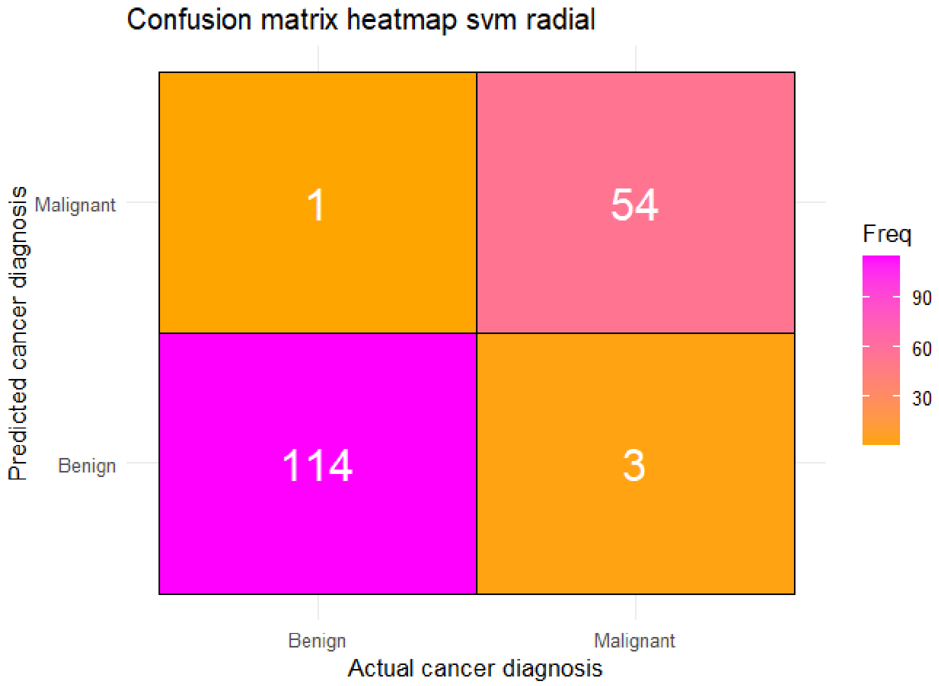


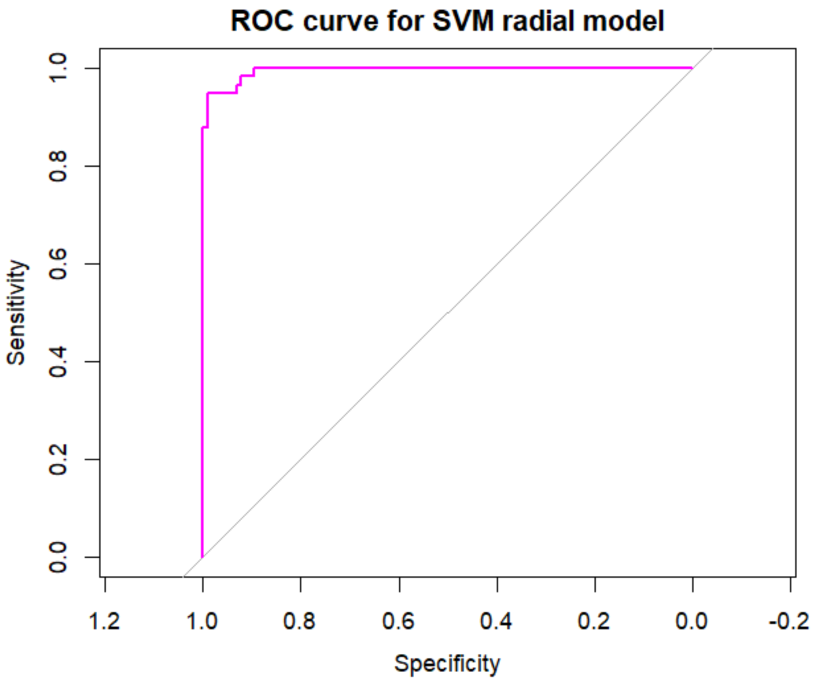
"Random forest accuracy: 0.936"

Area under the curve: 0.989

Support Vector Machines (SVM)

Support Vector Machines with radial, linear and polynomial basis functions were implemented, but the most accurate one among them was the one with the radial basis function:





"Svm radial accuracy: 0.9767"

Area under the curve: 0.995

Models' comparison

*From the AUC and Accuracy values and the respective ROC curve representations, we can say that the **Support Vector Machines (SVM) with radial basis function** is the most effective model for our classification task*

	Model	Accuracy
1	Logistic regression model	0.9535
2	Decision tree model	0.9186
3	Random forest model	0.9360
4	SVM radial model	0.9709
5	SVM linear model	0.9593
6	SVM polynomial model	0.9477

	Model	AUC
1	Logistic regression model	0.9946606
2	Decision tree model	0.9315027
3	Random forest model	0.9890160
4	SVM radial model	0.9949657
5	SVM linear model	0.9916095
6	SVM polynomial model	0.9948131